

Problem: Prove that the fraction $\frac{21n+4}{14n+3}$ is irreducible for every natural number n . (Source: 1959 IMO)

Solution: According to Bezout's identity, $21n+4$ and $14n+3$ are relatively prime if and only if there exist integers x and y such that $(21n+4)x + (14n+3)y = 1$.

Let $x = -2$ and $y = 3$.

$$\begin{aligned}(21n+4)x + (14n+3)y &= (21n+4)(-2) + (14n+3)(3) \\ &= -42n - 8 + 42n + 9 \\ &= 1\end{aligned}$$

It follows, from Bezout's identity, that $21n+4$ and $14n+3$ are relatively prime.

Thus the fraction $\frac{21n+4}{14n+3}$ is irreducible for every natural number n .